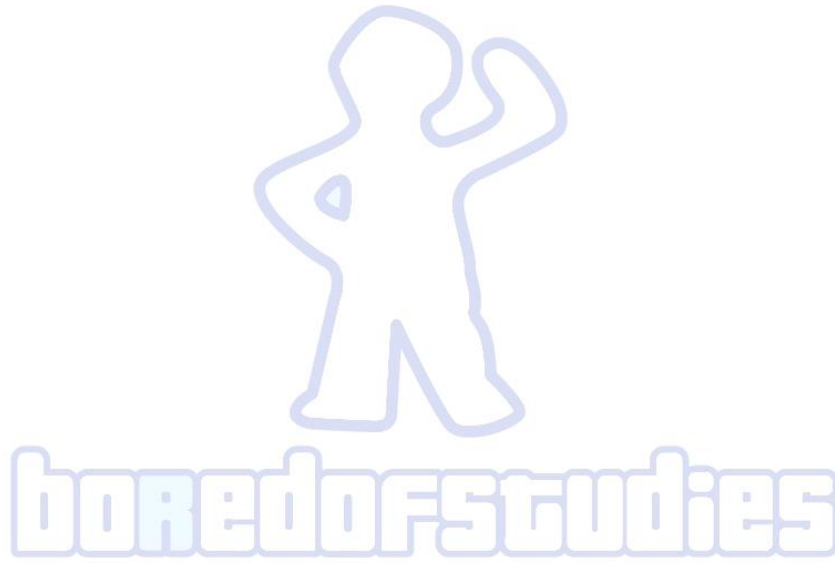




2019 Bored of Studies Trial Examinations

# Mathematics Extension 2

10<sup>th</sup> October 2019

## General Instructions

- Reading time – 5 minutes.
- Working time – 3 hours.
- Write using black or blue pen. Black pen is preferred.
- NESA-approved calculators may be used.
- A reference sheet has been provided.
- Show all necessary working in Questions 11 – 16.

## Total Marks – 100

**Section I** Pages 2 – 4

### 10 marks

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section.

**Section II** Pages 5 – 17

### 90 marks

- Attempt Questions 11 – 16
- Allow about 2 hours 45 minutes for this section.

## Section I

10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10.

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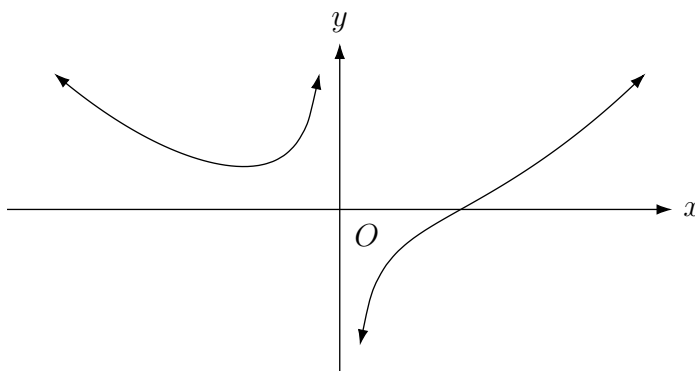
- 1 A relation has the implicit derivative  $\frac{dy}{dx} = xy$ . Which of the following is  $\frac{d^2y}{dx^2}$ ?
- (A)  $y$  (B)  $y + x$  (C)  $y(x^2 + 1)$  (D)  $x(y^2 + 1)$
- 2 An object with mass  $m$  falls vertically downwards and experiences a resistance force of  $kv$  where  $k$  is a constant and  $v$  is the particle's velocity at time  $t$ . Let  $g$  be the acceleration due to gravity. Which of the following is a possible acceleration equation of the object?
- (A)  $g - kv$  (B)  $-g - kv$  (C)  $g + \frac{k}{m}v$  (D)  $-g + \frac{k}{m}v$

- 3 For positive integers  $m$  and  $n$ , consider the complex number

$$z = \cos\left(\frac{(2m-1)\pi}{n}\right) + i \sin\left(\frac{(2m-1)\pi}{n}\right)$$

where  $2m-1$  and  $n$  have no common factors other than 1. The complex numbers  $z^k$  are plotted on an Argand diagram for all integers  $k > 0$ . How many distinct points are there?

- (A)  $n$  (B)  $2n$  (C)  $m$  (D)  $2m$
- 4 The following shows a sketch of  $y = \frac{x^k - 1}{x}$ .



Which of the following is a possible value for  $k$ ?

- (A) 0 (B) 1 (C) 2 (D) 3

- 5 A particle moves in circular motion along the unit circle  $x^2 + y^2 = 1$  with an angular displacement of  $\theta$  at time  $t$  to the positive  $x$ -axis. It is initially at rest at  $\theta = 0$ , before accelerating such that  $\ddot{\theta} = 1$ . Which of the following expressions best represents the acceleration of the particle in the  $y$  direction?

(A)  $\cos \theta - 2\theta \sin \theta$  (B)  $-(\cos \theta + 2\theta \sin \theta)$

(C)  $\sin \theta - 2\theta \cos \theta$  (D)  $-(\sin \theta + 2\theta \cos \theta)$

- 6 Let  $z_1$  and  $z_2$  be complex numbers such that  $z_1$  is in the first quadrant and  $z_2$  is in the third quadrant on the Argand diagram. Which of the following statements is correct?

(A)  $\frac{z_1}{z_2}$  is in the first or second quadrant (B)  $\frac{z_1}{z_2}$  is in the second or third quadrant

(C)  $\frac{z_1}{z_2}$  is in the third or fourth quadrant (D)  $\frac{z_1}{z_2}$  is in the fourth or first quadrant

- 7 How many distinct points of intersection are there between the curves described by the following relations?

$$\frac{x^2}{9} + \frac{y^2}{4} = 1 \quad \text{and} \quad (x+1)^2 - y^2 = 4$$

(A) 1 (B) 2 (C) 3 (D) 4

- 8 Let  $P(x) = c_0 + c_1x + c_2x^2$  with non-zero real coefficients  $c_0, c_1$  and  $c_2$ . Consider the following fraction for positive values of  $m$  and  $n$

$$\frac{P(x)}{(x+m)^2(x^2-n^2)}.$$

Which of the following could be equivalently expressed as the above fraction for some real numbers  $a, b$  and  $c$ ?

(A)  $\frac{a}{x+m} + \frac{b}{(x+m)^2} + \frac{c}{x^2-n^2}$

(B)  $\frac{a}{(x+m)^2} + \frac{b}{x+n} + \frac{c}{x-n}$

(C)  $\frac{a}{(x+m)^2} + \frac{b}{x+n} + \frac{c}{x^2-n^2}$

(D)  $\frac{a}{(x+m)^2} + \frac{b}{x-n} + \frac{c}{x^2-n^2}$

- 9 Which of the following is equivalent to the definite integral below?

$$\int_0^{\frac{\pi}{2}} \frac{1}{1 + \sin x} dx$$

(A)  $\int_0^{\frac{\pi}{2}} \frac{1 + \sin x}{1 + \cos x} dx$

(B)  $\int_0^{\frac{\pi}{2}} \frac{1}{1 + \left(\sin \frac{x}{2} + \cos \frac{x}{2}\right)^2} dx$

(C)  $\int_0^1 \frac{2}{(1+x)^2} dx$

(D)  $\int_0^1 \frac{1}{x^2 - 2} dx$

- 10 What is the general solution to the following system of equations for  $k = 1, 2, 3, \dots, n$ ?

$$2x_1 + x_2 + x_3 + \cdots + x_n = 1$$

$$x_1 + 2x_2 + x_3 + \cdots + x_n = 2$$

$$x_1 + x_2 + 2x_3 + \cdots + x_n = 3$$

$$\vdots$$

$$x_1 + x_2 + x_3 + \cdots + 2x_n = n$$

(A)  $x_k = k - \frac{n}{2}$

(B)  $x_k = k + \frac{n}{2}$

(C)  $x_k = k - \frac{n(n+1)}{2}$

(D)  $x_k = k + \frac{n(n+1)}{2}$

## Section II

**90 marks**

**Attempt Questions 11—16**

**Allow about 2 hours and 45 minutes for this section**

Answer each question in the appropriate writing booklet. Extra writing booklets are available. Throughout Questions 11–16, your responses should include relevant mathematical reasoning and/or calculations.

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**Question 11** (15 marks) Use the Question 11 Writing Booklet

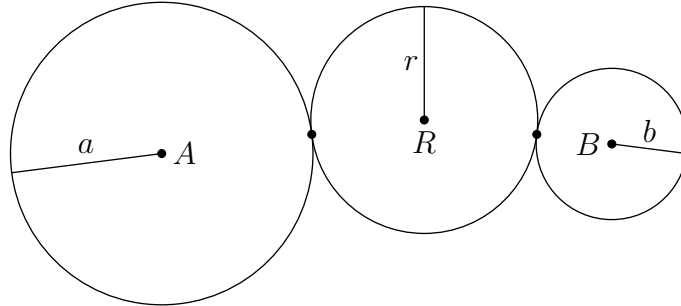
- (a) Evaluate  $\int_0^1 \frac{\tan^{-1}(\sqrt{x}) + \tan^{-1}(x^2)}{1+x^2} dx$ . **3**
- (b) Suppose that a polynomial  $P(x)$  with real coefficients has a non-real root  $\alpha$  of multiplicity  $r \geq 2$ .
- (i) Express  $P(x)$  as a product of real factors and some real polynomial  $Q(x)$  in terms of  $x$ ,  $\alpha$  and  $r$ . **1**
- (ii) Hence, show that  $P'(\alpha) = 0$ . **2**
- (c) (i) Sketch the graphs of  $y = x^3 + 1$  and  $y = x^2$  on the same set of axes. **1**
- (ii) Hence, sketch the graph of  $y = x^3 + x^2 + 1$ . **2**
- (iii) The region enclosed by the curve  $y = x^3 + x^2 + 1$ , the line  $x = -1$  and the coordinate axes is rotated about the  $y$ -axis. Find the volume of the solid of revolution formed. **2**
- (d) Let  $P(x) = x^4 - 3x^3 - 4x^2 + 9x + 2$  have real roots  $\alpha, \beta, \gamma$  and  $\delta$ . By considering the factorisation of  $x^2 + 1$ , or otherwise, find  $(\alpha^2 + 1)(\beta^2 + 1)(\gamma^2 + 1)(\delta^2 + 1)$ . **2**

**Question 11 continues on page 6**

Question 11 (continued)

- (e) Consider two distinct non-overlapping circles with fixed centres  $A$  and  $B$  and fixed radii  $a$  and  $b$  respectively. Suppose that another circle with centre  $R$  and radius  $r$  is drawn such that it touches both circles.

2



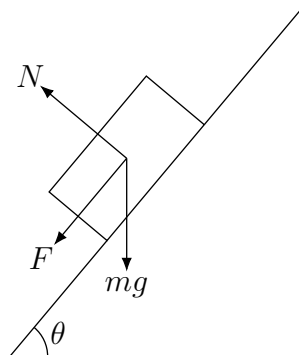
What is the geometric shape of the locus of the point  $R$  as  $r$  varies? Justify your answer. You do NOT need to provide any equations of this locus.

**End of Question 11**

**Question 12** (15 marks) Use the Question 12 Writing Booklet

- (a) An object of mass  $m$  moves along a circular track of radius  $r$  with constant speed  $v$  without slipping at time  $t$ . The track is banked at a positive acute angle of  $\theta$  to the horizontal.

The object experiences a normal force of  $N$  and a frictional force of  $F$  which is positive when directed down the banked track. Suppose that the friction force is such that  $0 \leq F \leq \mu N$  for some positive constant  $\mu$ . Let  $g$  be the acceleration due to gravity.



- (i) By resolving the forces horizontally and vertically, show that

**3**

$$v^2 \geq rg \tan \theta.$$

- (ii) Find the range of values of  $\mu$  in terms of  $\theta$ , such that

**2**

$$v^2 \leq \frac{rg(\tan \theta + \mu)}{1 - \mu \tan \theta}.$$

- (b) Let  $P_k \left( ct_k, \frac{c}{t_k} \right)$  for  $k = 1, 2, 3, 4$  be four points on the hyperbola  $xy = c^2$ . Suppose that the normals of  $P_1, P_2, P_3$  and  $P_4$  are concurrent and all intersect at the point  $R(a, b)$ . The equation of the normal at  $P_k$  is given by

$$t_k^3 x - t_k y - c(t_k^4 - 1) = 0 \quad (\text{Do NOT prove this}).$$

- (i) Show that  $t_1 t_2 t_3 t_4 = -1$ .

**1**

- (ii) Show that  $\frac{1}{t_1^2} + \frac{1}{t_2^2} + \frac{1}{t_3^2} + \frac{1}{t_4^2} = \frac{b^2}{c^2}$ .

**2**

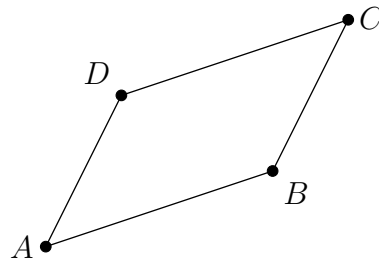
- (iii) Hence show that  $\frac{(P_1 R)^2 + (P_2 R)^2 + (P_3 R)^2 + (P_4 R)^2}{3} = a^2 + b^2$ .

**3**

**Question 12 continues on page 8**

Question 12 (continued)

- (c) Consider a parallelogram with vertices  $A, B, C$  and  $D$ .



- (i) Prove that  $AC^2 + BD^2 = 2(AB^2 + AD^2)$ . **2**
- (ii) Suppose that on the Argand diagram, the points  $A, B, C$  and  $D$  represent the complex numbers  $-z, 1, z$  and  $-1$  respectively. **2**

Use the result in (i) to describe the locus traced by  $z$  if  $\left|z - \frac{1}{z}\right| = 2$ .

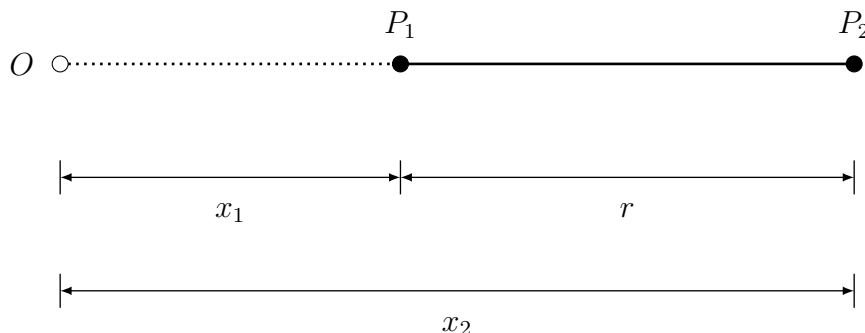
You do NOT need to specify the Cartesian equation of this locus.

**End of Question 12**



**Question 13** (15 marks) Use the Question 13 Writing Booklet

- (a) Two particles,  $P_1$  and  $P_2$  have masses  $m_1$  and  $m_2$  respectively, and are both initially at rest. At time  $t$ , they are  $r$  units apart.  $P_1$  and  $P_2$  exert forces on each other, so that  $P_1$  moves to the right towards  $P_2$ , and  $P_2$  moves to the left towards  $P_1$ . Let  $x_1$  and  $x_2$  be the displacements of  $P_1$  and  $P_2$  from a fixed point  $O$ .



The forces that  $P_1$  and  $P_2$  experience are given by the following equations

$$F_1 = \frac{Gm_1m_2}{r^2} \quad \text{and} \quad F_2 = -\frac{Gm_1m_2}{r^2} \quad (\text{Do NOT prove this}),$$

where  $G > 0$  is constant. Let  $r_0$  be the initial distance between the two particles.

(i) Explain why  $\ddot{x}_1 = \frac{Gm_2}{r^2}$  and  $\ddot{x}_2 = -\frac{Gm_1}{r^2}$ . **1**

(ii) Let  $M$  be the combined masses of the two particles. Show that  $\ddot{r} = -\frac{GM}{r^2}$ . **1**

(iii) Show that  $\dot{r} = -\sqrt{2GM \left( \frac{1}{r} - \frac{1}{r_0} \right)}$ . **2**

(iv) Hence show that for  $0 < r < r_0$  **3**

$$t = \frac{r_0\sqrt{r_0}}{\sqrt{2GM}} \left( \frac{\pi}{2} + \frac{1}{2} \sin 2\theta - \theta \right)$$

where  $\theta = \sin^{-1} \sqrt{\frac{r}{r_0}}$ .

(v) Using the fact that  $\sin x < x$  for positive values of  $x$  (do NOT prove this) show that **2**

$$t < \frac{\pi}{\sqrt{GM}} \left( \frac{r_0}{2} \right)^{\frac{3}{2}}.$$

Explain the physical significance of this result.

**Question 13 continues on page 10**

Question 13 (continued)

(b) By considering the expression

**2**

$$1 + (1 + x) + (1 + x)^2 + \dots + (1 + x)^{n-1}$$

show that for  $x \geq -1$  and positive integer  $n$

$$(1 + x)^n \geq 1 + nx.$$

(c) Suppose that  $x_1, x_2, \dots, x_n$  are positive real numbers. Let

$$a_n = \frac{x_1 + x_2 + \dots + x_n}{n}.$$

(i) Using the result in part (b), show that  $\left(\frac{a_{n+1}}{a_n}\right)^{n+1} \geq \frac{x_{n+1}}{a_n}$ .

**1**

(ii) Hence prove by mathematical induction that for positive integers of  $n$

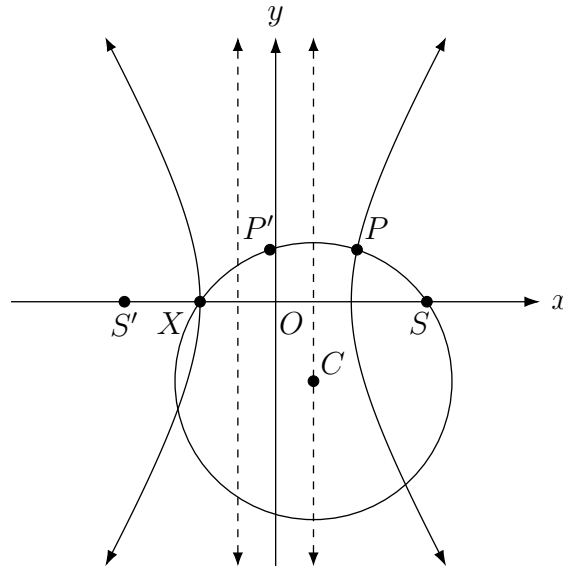
**3**

$$x_1 x_2 \dots x_n \leq \left(\frac{x_1 + x_2 + \dots + x_n}{n}\right)^n.$$

**End of Question 13**

**Question 14** (15 marks) Use the Question 14 Writing Booklet

- (a) The hyperbola  $\frac{x^2}{a^2} - \frac{y^2}{3a^2} = 1$  with foci  $S$  and  $S'$  is shown in the diagram below. Let  $X$  be the  $x$ -intercept of the hyperbola on the negative  $x$ -axis.



Let  $C$  be a point on the positive directrix and in the fourth quadrant such that it is the centre of a circle passing through the points  $S$  and  $X$ . Let  $P$  be the point of intersection of the circle and the hyperbola in the first quadrant. Let  $P'$  be the reflection of  $P$  about the positive directrix.

- |       |                                                        |          |
|-------|--------------------------------------------------------|----------|
| (i)   | Find the eccentricity of the hyperbola.                | <b>1</b> |
| (ii)  | Prove that $\triangle SCP \equiv \triangle XCP'$ .     | <b>2</b> |
| (iii) | Hence, show that $\angle SCX = 3 \times \angle PCP'$ . | <b>3</b> |

**Question 14 continues on page 12**

Question 14 (continued)

- (b) Let  $P(x)$  be a monic polynomial of the form

$$P(x) = x^4 - ax^3 - bx^2 - cx - d$$

where  $a \geq b \geq c \geq d > 0$  are integers. Let  $P(x)$  have non-zero roots  $\alpha$ ,  $\beta$ ,  $\gamma$  and  $\delta$ .

- (i) Show that an equation with the roots  $\frac{1}{\alpha}$ ,  $\frac{1}{\beta}$ ,  $\frac{1}{\gamma}$  and  $\frac{1}{\delta}$  is 1

$$dx^4 + cx^3 + bx^2 + ax - 1 = 0.$$

- (ii) Suppose  $P(x)$  has a positive integer root  $x = n$ . Using part (i), show that 1  
 $n > a$ .

- (iii) Hence, by finding a contradiction, show that  $P(x)$  cannot have a positive 2  
integer root.

- (c) Let  $\omega$  be a root of  $z^n = -1$ .

- (i) Show that 1

$$\frac{1}{1-\omega} + \frac{3}{1-\omega^3} + \frac{5}{1-\omega^5} + \dots + \frac{2n-1}{1-\omega^{2n-1}} = \sum_{k=1}^{2n-1} \frac{k}{1-\omega^k} - \sum_{k=1}^{n-1} \frac{2k}{1-\omega^{2k}}.$$

- (ii) Hence show that 1

$$\sum_{k=1}^{2n-1} \frac{k}{1-\omega^k} - \sum_{k=1}^{n-1} \frac{2k}{1-\omega^{2k}} = \sum_{k=n}^{2n-1} \frac{k}{1-\omega^k} - \sum_{k=1}^{n-1} \frac{k}{1+\omega^k}.$$

- (iii) Deduce that 3

$$\frac{1}{n} \left[ \frac{1}{1-\omega} + \frac{3}{1-\omega^3} + \dots + \frac{2n-1}{1-\omega^{2n-1}} \right] = 1 + \frac{1}{1+\omega} + \frac{1}{1+\omega^2} + \dots + \frac{1}{1+\omega^{n-1}}.$$

**End of Question 14**

**Question 15** (15 marks) Use the Question 15 Writing Booklet

- (a) A triangle has side lengths  $a$ ,  $b$  and  $c$  where  $a \geq b \geq c$ . Show that

**3**

$$\sqrt[3]{\frac{a^3 + b^3 + c^3 + 3abc}{2}} > a.$$

- (b) Let  $I_n = \int_0^{\frac{\pi}{4}} \tan^n x \, dx$  for integers  $n \geq 0$ .

(i) Show that  $I_n + I_{n-2} = \frac{1}{n-1}$  for  $n \geq 2$ .

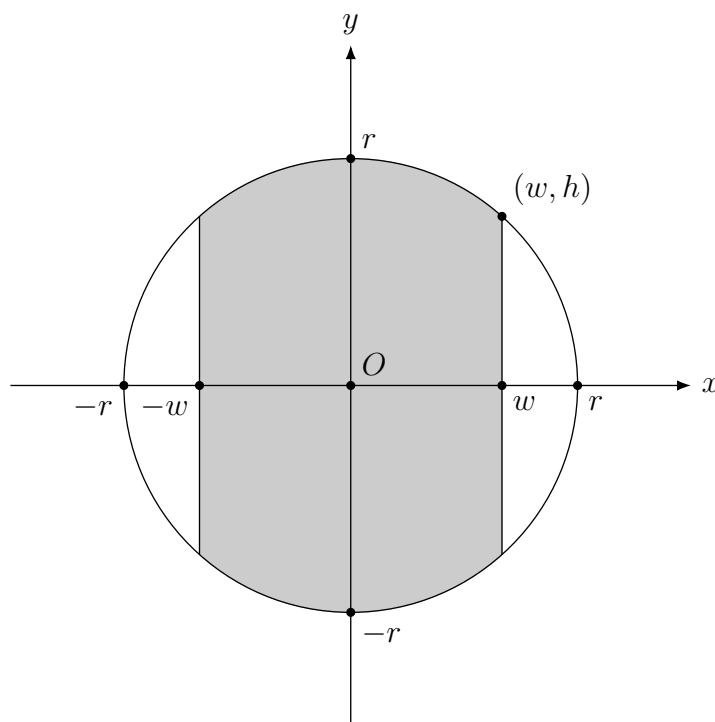
**2**

- (ii) Hence, evaluate  $I_6$ .

**1**

- (c) The diagram below shows the area bounded by the circle  $x^2 + y^2 = r^2$  and the lines  $x = w$  and  $x = -w$  for some positive values of  $r$  and  $w$ .

**2**



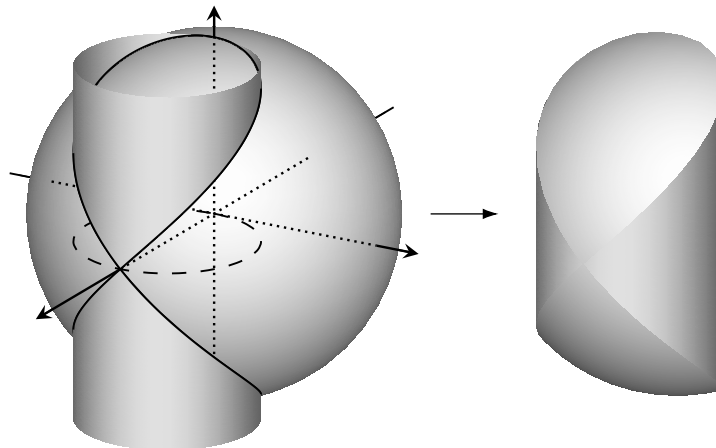
Let  $(w, h)$  be a point on the circle as shown in the diagram for some positive value of  $h$ . By considering, areas of sectors, or otherwise, show that the shaded area is given by

$$A = 2(w^2 + h^2) \tan^{-1} \left( \frac{w}{h} \right) + 2hw.$$

**Question 15 continues on page 14**

Question 15 (continued)

- (d) Consider the solid formed by the intersection of a cylinder of radius  $\frac{1}{2}$  units and a sphere of radius 1 unit as shown in the diagram below.



Vertical slices are taken as shown in the diagram below.



Let  $s$  be distance between the centre of the slice to the centre of the sphere.  
Let  $h, w$  and  $r$  be defined similar to that in part (c).

- (i) Show that  $h^2 + s^2 + w^2 = 1$ . 1
- (ii) Explain why  $\left(s - \frac{1}{2}\right)^2 + w^2 = \frac{1}{4}$  1
- (iii) Hence, show that the area of the slice is given by 2  

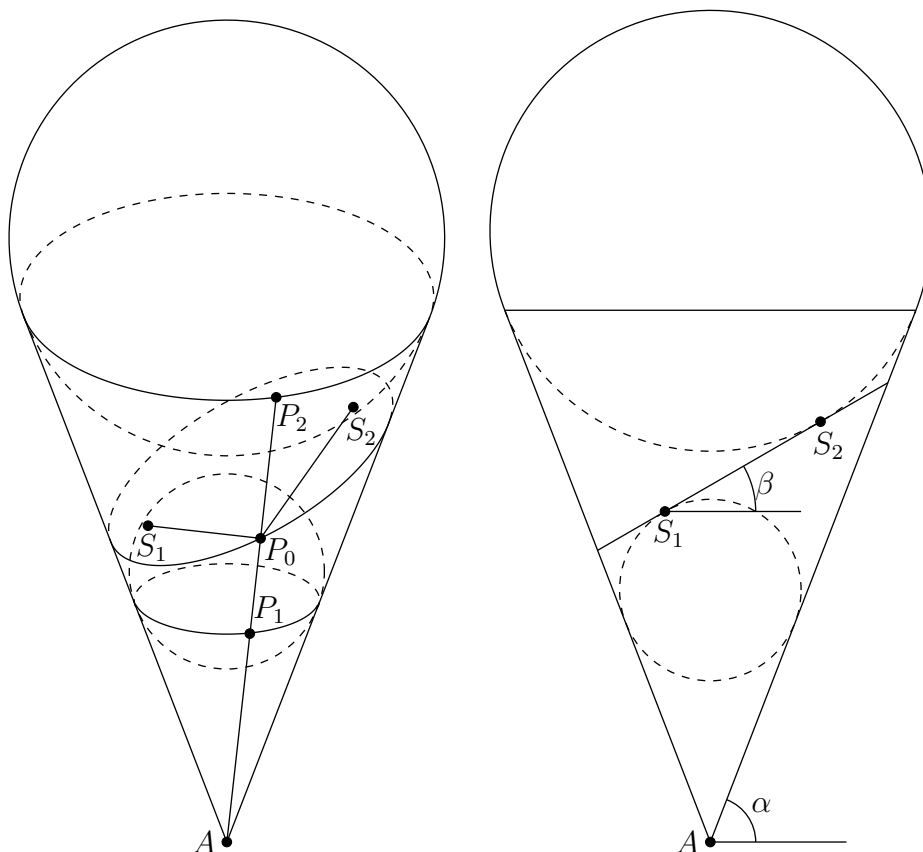
$$A = 2(1 - s^2) \tan^{-1} \sqrt{s} + 2\sqrt{s}(1 - s)$$
- (iv) Using the substitution  $\theta = \tan^{-1} \sqrt{s}$  and the result in question (b) (ii), or otherwise, find the volume of the solid. 3

**End of Question 15**

**Question 16** (15 marks) Use the Question 16 Writing Booklet

- (a) Consider a hollow right circular cone with vertex  $A$  which makes an angle of  $\alpha$  to the horizontal. A small sphere is inserted towards the vertex of the cone such that it touches the cone on the circle  $\mathcal{C}_1$ .

A flat ellipse is then inserted into the cone at an angle of  $\beta$  to the horizontal such that its edges touch the sides of the cone and one side of its surface touches the small sphere at  $S_1$ . Following that, a large sphere is then inserted to touch the ellipse at  $S_2$ . This sphere touches the cone on the circle  $\mathcal{C}_2$ .



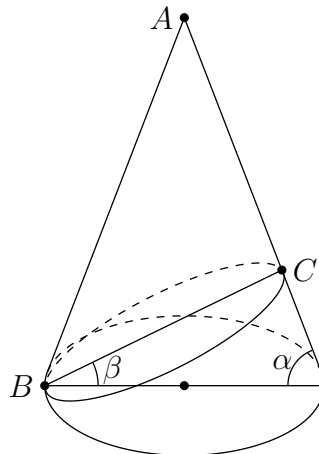
Let  $P_0$  be any point on the perimeter of the ellipse. The line  $AP_0$  is produced such that it touches the circle  $\mathcal{C}_1$  at  $P_1$  and  $\mathcal{C}_2$  at  $P_2$ .

- (i) Explain why  $P_0P_1 = P_0S_1$ . 1
- (ii) Hence deduce that  $S_1$  and  $S_2$  are the foci of the ellipse. 1

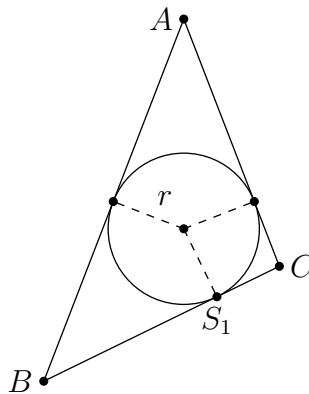
**Question 16 continues on page 16**

Question 16 (continued)

- (b) Consider a right circular cone with vertex  $A$  which makes an angle of  $\alpha$  to the horizontal. The cone is then sliced at point  $C$  to point  $B$ , on the base of the cone, at an angle of  $\beta$  to the horizontal, where  $\beta < \alpha$ . The cross-section of the slice is an ellipse with a semi-major axis of  $a$  and an eccentricity of  $e$ .



- (i) Show that  $AB = \frac{2a \sin(\alpha + \beta)}{\sin 2\alpha}$ . 1
- (ii) Show that  $AC = \frac{2a \sin(\alpha - \beta)}{\sin 2\alpha}$ . 1
- (iii) Consider a circle inscribed in  $\triangle ABC$  with radius  $r$  such that each side is tangent to the circle. Let the point of contact of  $BC$  with the circle be  $S_1$ . 1



Show that

$$r = \frac{AC \times AB \sin 2\alpha}{AB + BC + AC}.$$

**Question 16 continues on page 17**



Question 16 (continued)

- (iv) Using the result in part (a), show that **3**

$$e = 1 - \frac{r}{a} \cot \left( \frac{\alpha + \beta}{2} \right).$$

- (v) Hence, show that **3**

$$e = \frac{\sin \beta}{\sin \alpha}.$$

You may assume the following results without proof:

$$\sin \alpha + \sin \beta = 2 \sin \left( \frac{\alpha + \beta}{2} \right) \cos \left( \frac{\alpha - \beta}{2} \right),$$

$$\cos \alpha + \cos \beta = 2 \cos \left( \frac{\alpha + \beta}{2} \right) \cos \left( \frac{\alpha - \beta}{2} \right).$$

- (c) Student  $A$  and student  $B$  play a game where a fair coin is tossed repeatedly until one of them wins. Suppose that the rules of the game are as follows

- If there are 3 more heads than tails, student  $A$  wins the game.
- If there are 3 more tails than heads, student  $B$  wins the game.

- (i) What is the probability that student  $A$  wins the game? **1**

- (ii) Let  $p$  be the probability that student  $A$  wins the game, if the first toss was a head. Explain why **2**

$$p = \frac{1}{2} + \frac{1}{4} \times p.$$

- (iii) Suppose that the first toss was a tail instead. What is the probability that student  $A$  wins the game? **1**

**End of paper**